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Article

Apparent migration and natal homing of a small minnow in a large ancient lake

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Abstract

Migratory freshwater fish species are particularly susceptible to population decline because they rely on multiple habitats and open passages to complete their life cycle. Understanding the migratory patterns of cyprinids is thus essential for their management and conservation. *Gnathopogon caerelescens*, a small cyprinid species endemic to Lake Biwa (Japan), is one of the most important targets of inland fisheries. However, their abundance has dramatically decreased. By using an extensive alizarin complexone-mark-and-recapture program for of *G. caerelescens* juveniles, together with the collaboration of commercial fishermen, this cyprinid fish was demonstrated as: (1) migrating from a nursery lagoon toward the main lake and moving along the near-shore from spring to autumn; (2) migrating from the near-shore shallow habitats to off-shore deep habitats from autumn to winter; and (3) migrating to a nursery lagoon from off-shore deep habitats from winter to spring. These results highlighted the importance of protecting individual spawning/nursery habitats, and maintaining connectivity between each spawning/nursery habitat and the lake system.

Key words: cyprinids, conservation management, habitat protection, inland fisheries, migration, natal homing

Introduction

Of all the great spectacles of nature, animal migration is without doubt one of the most remarkable (Brönmark et al. 2013). Migration has been commonly defined as individuals or populations (or parts of populations) moving between two well-defined habitats on a temporally predictable basis (Brönmark et al. 2013). The three principal functional categories of migrations include reproductive (spawning) migration, refuge migration, and feeding migration (Northcote 1978, 1984). Reproductive migration involves the return or otherwise, of spawners to their birth place (natal homing) (Lucas and Baras 2001). If fish do not return to the birth place, they recognize spawning grounds aided by pheromonal odors released by conspecifics as described for the sea lamprey (*Petromyzon*) and also suggested the Arctic char (*Salvelinus*) (Bergstedt and Seelye 1995; Sorensen et al. 2005; Nordeng 2009). Whereas the ability to home to a particular natal spawning location after migrations of thousands of kilometers has been well documented for anadromous salmonid species, following mark and recapture studies (Dittman and Quinn 1996; Lucas and Baras 2001), migration in most freshwater fishes has been thought to be relatively rare, due to most species being considered as largely sedentary and undergoing only limited local movements (Lucas and Baras 2001; Brönmark et al. 2013). However, advances in technology for tracking individuals has led to a reassessment of the prevalence of migratory behavior in freshwater fishes (Lucas and Baras 2001; Chapman et al. 2011, 2012*a*, 2012*b*; Brönmark et al. 2013), many species undertaking spawning migrations in the river systems (Tynus 1990; Lucas and Batley 1996; Irving and Moddle 2000; Rakowitz et al. 2008). In addition, refuge migration from adverse environmental factors, such as low temperature, oxygen and predation, has also been demonstrated (Jepsen and Berg 2002; Skov et al. 2008, 2011). Finally, feeding migrations are common when food resources fluctuate in a predictable manner across time and space (Tynus 1991; Brönmark et al. 2013). Although each migratory pattern (spawning, refuge, and feeding migration) of freshwater fishes has been well studied, none have investigated freshwater fish utilization of feeding, refuge and spawning habitats on spatial and temporal scales throughout the fish life history. This may underline the subsequent decline of freshwater fishes, since migratory freshwater fish species are particularly susceptible

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to population decline, as they rely on multiple habitats and open passages to destinations to complete their life cycle (Lucas and Baras 2001; Wilcove and Wikelski 2008; Chapman et al. 2012*a*, 2012*b*).

The widely distributed freshwater family Cyprinidae (hereinafter cyprinids) is often the major component of lake and river fish communities (Winfield and Nelson 1991; Stnout et al. 2016). Cyprinids comprise many species with varying size, life histories and ecology, and have been the subject of a number of migration studies, primarily in Europe and North America (Lucas and Batley 1996; Chapman et al. 2012*a*, 2012*b*; Walters et al. 2014; Banet et al. 2022, but see Sungani et al. 2016). However, although cyprinids are an important target fish for Asian inland fisheries (Bnińsla 1991; Allan et al. 2005; Kano et al. 2016), migration studies of the former have been limited in Asia (Kunimune et al. 2011). Honmoroko (*Gnathopogon caerulescens*), a small cyprinid fish endemic to Lake Biwa, an ancient lake in central Japan, is one of the most important target species of inland fisheries in Lake Biwa. The fish had been previously harvested at approximately 150–350 tons per year, but the catch declined drastically in the mid-1990s. Since 1992, early summer water levels in the lake have been heavily reduced by water management measures to minimize the risk of downstream flooding (Kikko et al. 2012; Fujioka 2013). This suggested that desiccation of eggs by the water level control had caused the decline of the species (Kikko et al. 2012; Fujioka 2013). Honmoroko exhibits an exclusive pelagic lifestyle in the lake foraging for zooplankton, and usually live for just a single year (Nakamura 1949; Fujioka and Saegusa 2015). Honmoroko has been thought to undertake the following annual migration cycle: (1) adult migration from the main lake to the lagoon for spawning in spring (Kikko et al. 2015*a*); (2) juvenile migration from the nursery lagoon to a pelagic habitat of the main lake for feeding in summer (Kikko et al. 2018*a*); and (3) subadult and adult migration in the main lake from near-shore shallow areas to offshore deep areas in autumn and winter (Maki 1966; Fujioka and Saegusa 2015). However, few studies have been conducted to clarify: (1) whether all juveniles migrate from their nursery lagoon habitats to the main lake (i.e., some fish remain as residents in the lagoon; partial migration, Nakamura 1949) (but see, Kikko et al. 2018*a*); (2) the spatial extent to which subadult and adult fish migrate within the main lake; and (3) whether matured adults return to their natal lagoon for spawning. Therefore, from the viewpoint of conservation/management for this small migratory cyprinid, it is necessary to evaluate these key migration features during juvenile to adult stages.

Studies of migration have been hindered by the difficulties of conducting mark-recapture studies of species that are characterized by the production of large numbers of small juveniles that suffer from high initial mortality (cf. Thorrold et al. 2001). Standard fish marking methods, such as tagging and fin clipping, are often limited by the size and quantity of fishes to be marked (Barker and Mckaye 2004). However, immersion marking of fish otoliths with fluorescent dyes, such as the alizarin complexone (ALC), is useful for marking a huge number of small fish, being cost-effective and having a higher retention rate over time (Lü et al. 2017). In addition, double and single ALC marked fish have been correctly identified in both wild and captive conditions (Tsukamoto et al. 1989; Fujiwara et al. 2010; Lü et al. 2017; Kikko et al. 2019). For this study, mass ALC marking of fish combined with the high production of juveniles utilizing rice fish cultivation skills (Kikko et al. 2013; Taguchi et al. 2015) was believed to have great potential in mark-recapture, migration studies during juvenile to adult stages. This study tested the hypothesis that honmoroko move out of nursery lagoons each spring to enter Lake Biwa, and then return to their natal lagoon to spawn the following spring.

Materials and methods

Study area

Lake Biwa, the largest lake in Japan (674 km^2) and one of the world's few ancient lakes, comprises a large deep northern basin (617.8 km^2, mean depths 43 m) and small shallow southern basin (52.5 km^2, 4 m). Nishinoko Lagoon (2.2 km^2, 1.5 m mean depth), Ibanaiko Lagoon (0.5 km^2, 1.5 m), Matsunoki Lagoon (0.2 km^2, 1.5 m), and Sonenuma Lagoon (0.2 km^2, 1.2 m) around Lake Biwa are all connected to the main lake (Fig. 1). Currently, honmoroko spawn mostly in the lagoons adjacent to the northern basin and along western and eastern shoreline habitats in northern parts of the northern basin (Fig. 6, Kikko et al. 2015*a*, 2015*b*, 2015*c*, 2015*d*). Nishinoko Lagoon and Ibanaiko Lagoon were selected as the stocking sites for ALC-marked fish, due to their importance as both spawning and nursery habitats for wild honmoroko (Fig. 1; Kikko et al. 2014, 2015*a*, 2015*b*, 2015*c*). The outlet of Nishiniko lagoon (Chomeiji River: 80 m in width, 4 km in length, and 2 m in mean depth) is connected to the main lake by a mostly stagnant water flow (Fig. 1). Rice fields located near the southern foreshore of Nishinoko Lagoon are connected to the lagoon via agricultural ditches (Fig. 1). Similarly, the outlet of Ibanaiko Lagoon (Daido river: 120 m in width, 3 km in length, and 2 m in mean depth) to the main lake is a mostly stagnant water flow. Weirs in both the Chomeiji and Daido rivers, constructed near the outlet mouths of their respective lagoons (Kikko et al. 2018*a*), are often closed so as to maintain the water level in the lagoons for agricultural use, depending on requirements.

Mass ALC mark-and-release program

On 18 June 2012, 694 700 ALC-marked juveniles were stocked into Nishinoko Lagoon to evaluate the migration during the juvenile to adult stages (hereinafter NL-population; Table 1). To test the reproducibility of the results, 774 100 and 128 000 ALC-marked juveniles were stocked into Nishinoko (NL-population 19 June 2013) and Ibanaiko (IL-population 24 May 2013) lagoons, using different ALC-markings for each lagoon thereby enabling subsequent identification of the lagoons of origin (Table 1; Kikko et al. 2018*a*). The brood stock for the NL- and IL-populations was sourced from major spawning habitats around Lake Biwa (western and eastern lakeshores), thereby excluding possible stock from Nishinoko and Ibanaiko lagoons (Kikko et al. 2016). One-day-old

Fig. 1. (*a*) Locations of Nishinoko, Ibanaiko, Matsunoki, and Sonemuma lagoons adjacent to Lake Biwa. (*b*) Locations of sampling stations around Nishinoko Lagoon, its outlet (Chomeiji River) (Stations 1–18) and rice fields. Star, release point of alizarin complexone (ALC)-marked individuals produced in rice fields through ditches to Nishinoko Lagoon. Numbers on maps indicate isobaths depths.

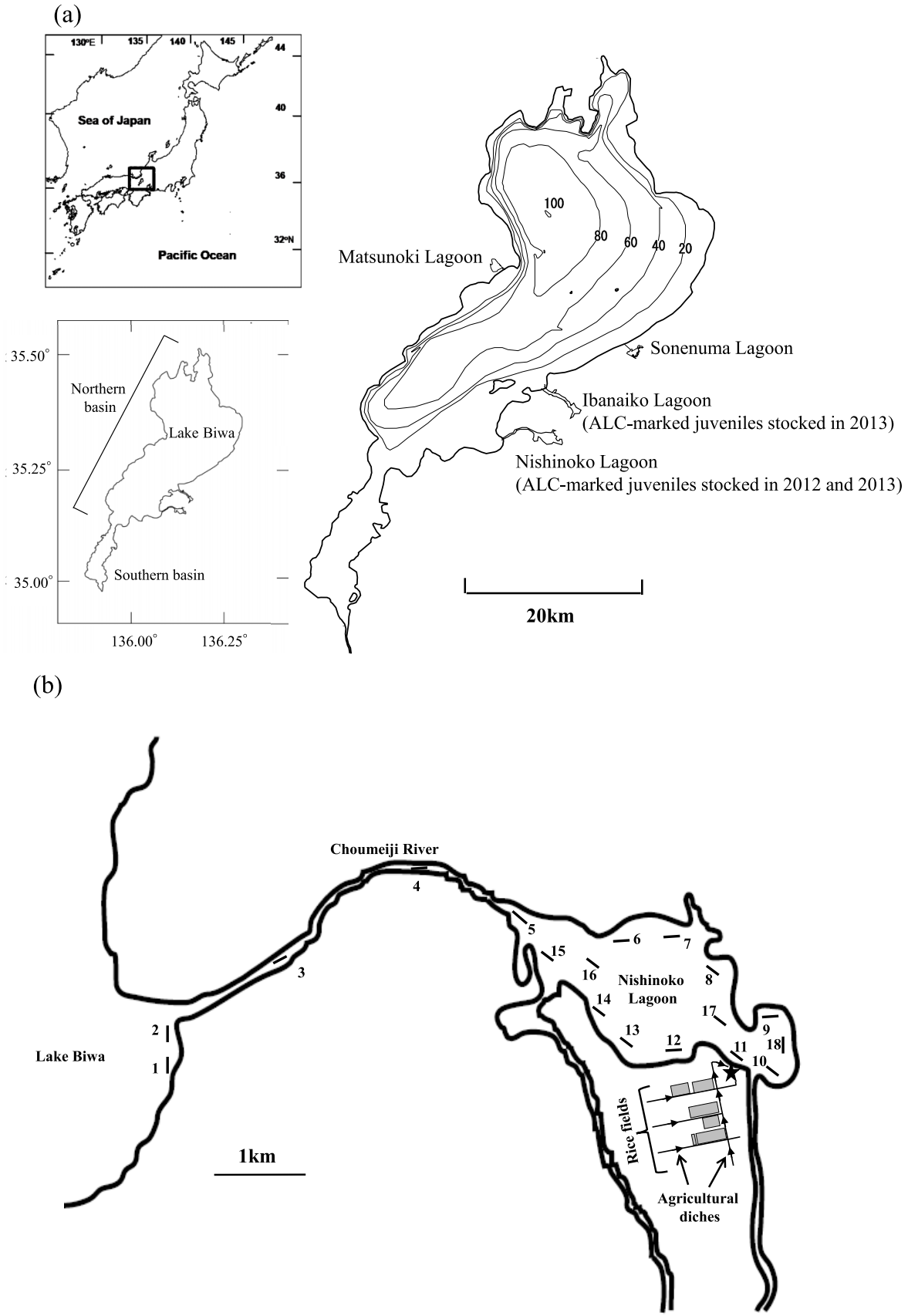

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Table 1. Size and number of alizarin complexone (ALC)-marked honmoroko stocked to Nishinoko and Ibanaiko lagoons, and Lake Biwa recapture data.

Date	Sampling category[a]	No. of fish[b]	ALC marks[c]	Sex	Standard length (mm) mean ± SD	Distance from release point (km) mean ± SD
Nishinoko Lagoon						
18 June 2012	Stock	694 700	Small-ring (1)	–	22.6 ± 1.6	–
October–November 2012	Recapture	16 (3180)	Small-ring (1)	–	75.1 ± 2.4	14.9
January–February 2013	Recapture	78 (2554)	Small-ring (1)	Female	73.9 ± 3.6	27.0 ± 7.8
January–February 2013	Recapture	72 (2707)	Small-ring (1)	Male	71.5 ± 3.2	27.2 ± 7.5
March–June 2013	Recapture	49 (337)	Small-ring (1)	Female	73.3 ± 4.5 (11.9 ± 5.9)[d]	–
March–June 2013	Recapture	92 (1627)	Small-ring (1)	Male	70.9 ± 3.9 (7.2 ± 2.8)[d]	–
19 June 2013	Stock	774 100	Small-ring (1)	–	23.2 ± 1.8	–
October–November 2013	Recapture	130 (2556)	Small-ring (1)	–	81.3 ± 4.4	17.9 ± 5.8
January–February 2014	Recapture	222 (1984)	Small-ring (1)	Female	82.2 ± 4.4	20.7 ± 8.0
January–February 2014	Recapture	219 (2308)	Small-ring (1)	Male	77.2 ± 4.8	22.4 ± 9.6
March–June 2014	Recapture	49 (133)	Small-ring (1)	Female	80.8 ± 5.1 (11.1 ± 5.4)[d]	–
March–June 2014	Recapture	68 (373)	Small-ring (1)	Male	77.2 ± 4.9 (9.7 ± 8.4)[d]	–
Ibanaiko Lagoon						
24 May 2013	Stock	128 000	Dot-ring (2)	–	16.8 ± 1.8	–
October–November 2013	Recapture	10 (2556)	Dot-ring (2)	–	83.7 ± 5.3	10.6 ± 3.8
January–February 2014	Recapture	9 (1984)	Dot-ring (2)	Female	81.4 ± 8.7	9.8 ± 4.5
January–February 2014	Recapture	10 (2308)	Dot-ring (2)	Male	81.3 ± 8.7	15.2 ± 5.9
March–July 2014	Recapture	8 (378)	Dot-ring (2)	Female	90.9 ± 3.9 (12.5 ± 4.2)[d]	–
March–July 2014	Recapture	14 (357)	Dot-ring (2)	Male	88.6 ± 3.1 (6.6 ± 1.4)[d]	–

[a]ALC-marked fish released to Nishinoko or Ibanaiko lagoons; recaptures along Lake Biwa shoreline by gillnet, offshore by bottom trawl.
[b]ALC-marked stocked fish numbers approximate; total numbers of recaptured fish shown in parentheses.
[c]Numbers in parentheses indicate number of ALC rings or dots in otolith.
[d]Numbers in parentheses indicate gonadsomatic index.

honmoroko larvae cultured in water from Lake Biwa were soaked in ALC solution (16 mg l^{-1}) for 24 h to stain (small ring mark) the otoliths of individual fish (Fujiwara 1999). About 2 million ALC-marked honmoroko larvae were stocked into separate rice fields near the southern foreshore of Nishinoko Lagoon in late May 2012 and 2013 (Table 1; Fig. 1). These individuals were reared in the rice fields for about 30 days without any supplemental feeding, fertilizer, or chemical applications, following the methods described by Kikko et al. (2013) and Taguchi et al. (2015). In 2012 and 2013, 30 days after stocking, 694 700 and 774 100 ALC-marked juveniles (23 mm SL) were released into Nishinoko Lagoon (NL-population) by draining the rice fields through agricultural ditches (Table 1; Figs. 1 and 5).

Seasonal sampling

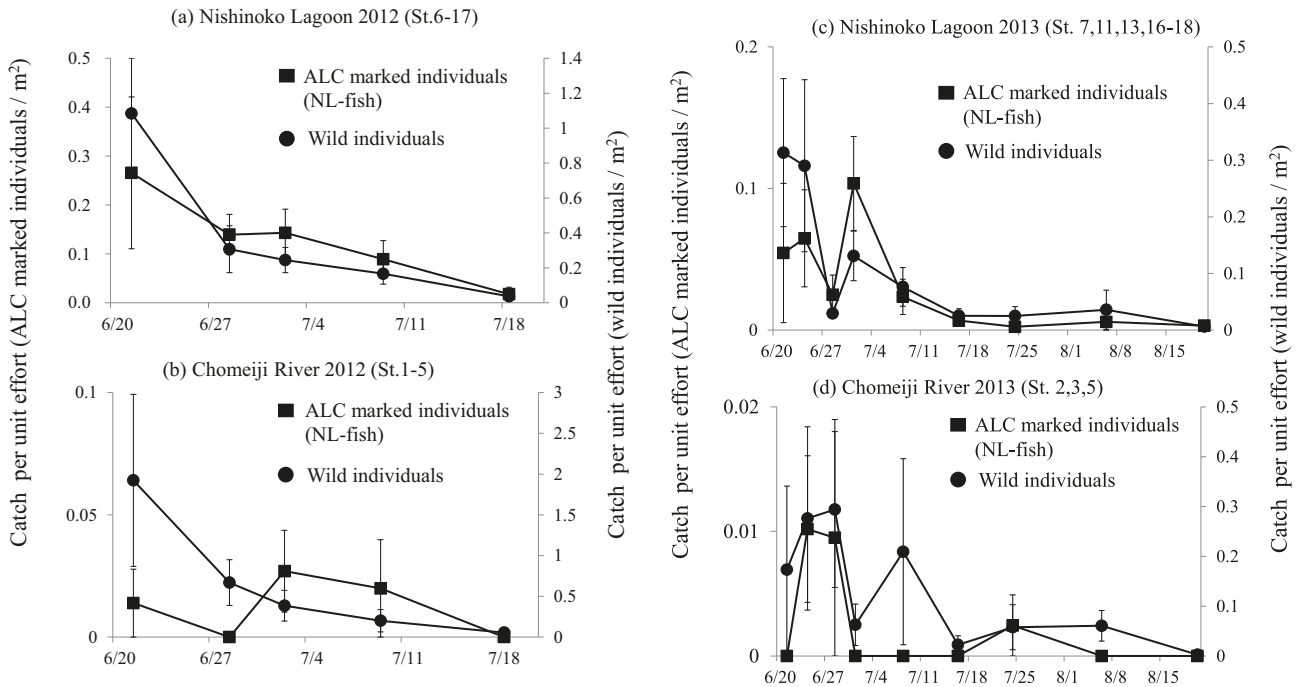
Migration patterns from nursery lagoons to lake habitats

To determine if honmoroko migrated from the natal habitat to the main lake, the migratory patterns of juveniles from Nishinoko Lagoon to Lake Biwa were surveyed in June and July in 2012, and from June to August in 2013, using a 3.0 m beam trawl (0.5 m height, 5 m overall length, 0.5 cm mesh cod end), towed by a boat (0.5 t). Surveys were conducted weekly during daytime (9:00–16:00), following the procedure described by Kikko et al. (2018*a*) (Fig. 1). Catch per unit effort (CPUE: captured fish (individuals)/seined area (m^2)) of ALC-marked and unmarked wild fish in the Nishinoko Lagoon (2012: St.6–17, 2013: 7, 11, 13, and 16–18; Fig. 2) and Chomeiji River (2012: St.1–5, 2013: St. 2, 3, and 5) were calculated for each sampling date, following Kikko et al. (2018*a*).

Lake wide migration across Lake Biwa

To quantify the migration of honmoroko across Lake Biwa during autumn and winter (just before homing migration), fish were obtained from commercial catches in October and November (hereafter, autumn) in 2012 (n = 3180) and 2013 (n = 2556) (Fig. 3), and in January and February (hereafter, winter) in 2013 (n = 5261) and 2014 (n = 4296) (Fig. 4). Fish were captured along the Lake Biwa foreshore by gill net (approximately 2.0 cm in mesh size) in autumn, and offshore in the lake by bottom trawl net (approximately 1.0 cm in mesh size) in winter, so as to track their seasonal habitat shift (Fujioka and Saegusa 2015). At the time, commercial catches by gill net or bottom trawl were extremely limited south of the release sites due to the reduced honmoroko distribution in that region in fall and winter, due to the absence of spawning in the southern basin since 1995 (Fujioka 2013). Consequently, due to sampling being dependent upon commercial fishermen, the sampling areas were restricted to north of the release sites (Figs. 3, 4). Date, location, and

Fig. 2. Changes in catch per unit effort (CPUE; mean ± S.E. (standard errors)) of alizarin complexone (ALC)-marked (NL-fish) and wild honmoroko individuals. (*a*) Nishinoko Lagoon, June and July 2012. (*b*) Chomeiji River, June and July 2012. (*c*) Nishinoko Lagoon, June and August 2013. (*d*) Chomeiji River, June and August 2013.



water depth for each fish captured were recorded by the fishermen.

Reproductive migration (natal homing)

To quantitatively evaluate the possibility of natal homing of honmoroko, fish were collected around the main spawning habitats throughout the March to early July spawning period in 2013 (n = 3646) and 2014 (n = 2247) (Fig. 6; Table 2). Fish were randomly captured by various methods, including commercial gillnet, boat electrofishing (Model 2.5GPP, Smith-Root, Inc., Vancouver, WA, USA), backpack electrofishing (Model 12, Smith-Root, Inc., Vancouver, WA, USA), set-net, bait fishing, and dip net (Table 2). Sampling effort by various methods in each spawning habitat ranged from 7 days to 23 days in 2013, and from 2 days to 33 days in 2014 (Table 2). It was not possible to ensure equal sampling site effort at all sites, because the range of sampling locations, sampling gear and collectors differed among the sites. Sampling was continued throughout the main spawning season in each spawning habitat, with the aim of achieving adequate representations of mature adult fish at each sampling location, and robust results.

Identification of ALC-marked fish

Otoliths of all captured fish were examined to determine the ALC-marked stock, based on the presence or absence of an ALC small-ring mark (NL-population) or ALC dot-ring mark (IL-population) in the otolith, using a fluorescent microscope (Nikon, OPTIPHOT, XT-EFD) with a G-filter (BM 520; www.nikon.com) (Table 1, Kikko et al. 2018*a*). Fish collected offshore in winter by bottom trawl were measured (standard length-SL) and their sex determined by dissection. Fish collected during the spawning season around the spawning habitats were measured (SL), weighted (W), sexed by dissection, and their gonads weighted. Gonadosomatic index (GSI = gonad weight × 100/body weight), a direct index of maturity condition, was also calculated (Kikko et al. 2014, 2015*a*). A condition factor ($W/SL^3 \times 10^6$, hereafter CF) was calculated for ALC-marked fish collected from the lakeshore in fall by gillnet and offshore in winter by bottom trawl. Honmoroko generally comprise almost exclusively age 0 individuals, with only a small percentage over age 1 year (determined by scale reading (Maki 1966; Fujioka and Saegusa 2015)). Because only a small percentage of specimens collected in the present study were over age 1, data analyses were restricted to the 2012 and 2013 (age 0) cohorts.

Data analyses

A generalized additive model (GAM; Hastie and Tibshirani 1986) (Binomial error distribution and logit-link function) was used to examine migratory patterns of honmoroko within the main lake in both the autumn and winter seasons, using MGCV packages (Wood 2017) implemented in R 3.4.3 (R Development Core Team 2017). In the GAM, the probability of ALC-marked fish at a sampling point was used as a dependent variable, fishing location (latitude and longitude), and sex being independent variables. Latitude and longitude were included as a smoothing function to test their possible nonlin-

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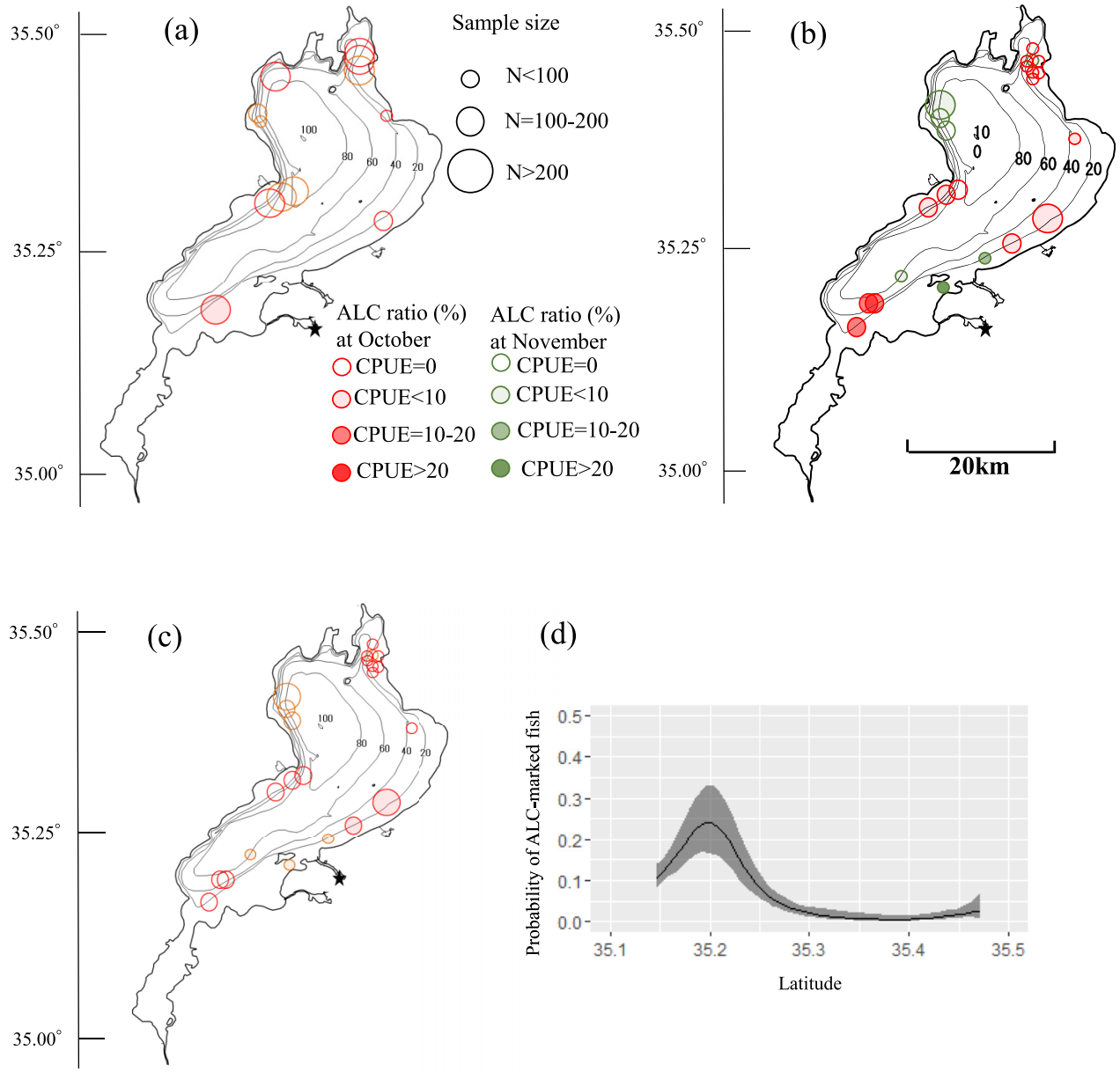
Fig. 3. Geographical distribution of alizarin complexone (ALC)-marked ratios of honmoroko in October and November. Stars, release points of ALC-marked individuals. (*a*) Stocked to Nishinoko Lagoon in June 2012: NL-fish. (*b*) Stocked to Nishinoko Lagoon in June 2013: NL-fish. (*c*) Stocked to Ibanaiko Lagoon in May 2013: IL-fish. Numbers on maps indicate isobaths depths. (*d*) Results of generalized additive model for probability of ALC-marked honmoroko in October and November, 2013; line represents median value; shaded area denotes 95% upper and lower credibility limits. CPUE, catch per unit effort.



ear effects on the probability of ALC-marked fish. Since, there were multiple sampling dates in each year (2012–2013 and 2013–2014), date and year of catch were used as random effects, implemented as "s (factor, bs="re")" in the GAM model. However, in the 2012 autumn survey, only 16 out of 3180 ALC-marked fish were captured at one sampling point on one occasion. Therefore, data for 2012 was omitted in favor of the 2013 data (with date as a random factor) for the analysis of autumn migratory patterns. In the initial model, all independent variables and their two-way interaction terms were included. The R package MuMIn (Barton 2009) was used to select the best possible combination of predictor variables contained within the GAM. Models were compared using a Bayesian information criterion (BIC; Schwarz 1978).

In the GAM models, latitude and longitude were used as independent variables to determine the directional movements from the release points (natal habitats) of ALC-marked fish to the recapture points within the main lake. An alternative variable might have been the direct distance from the release points of ALC-marked fish, but this could not be used to estimate the movement direction. Therefore, we did not include direct distance in the GAM models, although the movement distance from the release points of ALC-marked fish to recapture points within the main lake (direct distance from the

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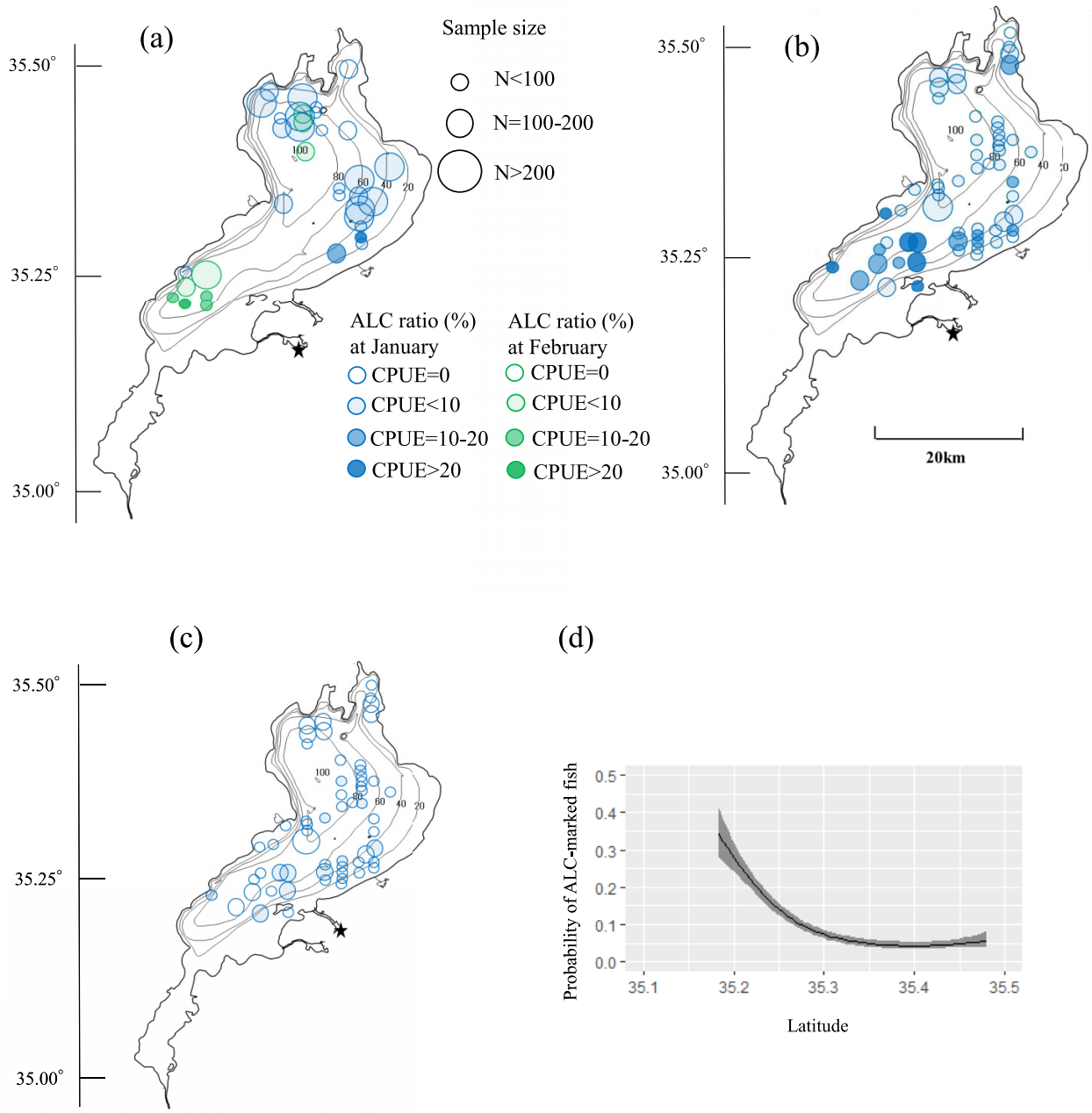
Fig. 4. Geographical distribution of alizarin complexone (ALC)-marked ratios of honmoroko in January and February. Stars, release points of ALC-marked individuals. (*a*) Stocked to Nishinoko Lagoon, June 2012: NL-fish. (*b*) Stocked to Nishinoko Lagoon, June 2013: NL-fish. (*c*) Stocked to Ibanaiko Lagoon, May 2013: IL-fish. Numbers on maps indicate isobaths depths. (*d*) Results of generalized additive model for probability of ALC-marked honmoroko in January and February, 2013 and 2014; line represents median value; shaded area denotes 95% upper and lower credibility limits. CPUE, catch per unit effort.



lagoon outlet river mouths to the recapture points of ALC-marked fish within the main lake, in addition to distance from the release points to the lagoon outlet river mouths (following the stream network)) were separately calculated.

To understand the propensity for natal homing, differences in ratios of ALC-marked fish among the spawning habitats were analyzed using χ^2 test, followed by Ryan multiple comparison tests. The homing ratio (return rates) of the NL-population in each year was calculated as follows, using data obtained during the spawning season; (ALC small ring-marked fish caught in Nishinoko Lagoon)/(Total ALC small ring-marked fish caught from all of the spawning habitats) × 100 (%) (Hendry et al. 2004). The homing ratio of the IL-population was similarly calculated.

Results

Migration from nursery to lake habitats

The CPUE of the NL-population in the Nishinoko Lagoon peaked in late June, but had decreased to less than 10% of that level by late July in both 2012 and 2013 (Figs. 2*a* and 2*c*). The

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Table 2. Results of model selection for generalized additive model analyses of migration pattern of honmoroko (*Gnathopogon caerelescens*) in Lake Biwa in autumn and winter.

Model	Sex	s (Latitude)	s (Longitude)	s (Lat × Long)	s (Lat × sex)	s (Long × sex)	BIC	Δ BIC
Autumn season (October–November)								
1		+					982.2	0
2		+	+				995.7	13.48
3			+				1016.3	34.10
Winter season (January and February)								
1		+					4060.8	0
2		+	+				4062.8	2.06
3		+			+		4064.8	4.09
4					+		4078.3	17.58

Note: Probability of alizarin complexone-marked honmoroko at a sampling point used as a dependent variable; capture location (latitude and longitute) and sex used as independet variables. BIC, Bayesian information criterion.

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CPUE of the NL-population in the Chomeiji River (connecting Nishinoko Lagoon and the main lake) also peaked in late June, but had decreased to zero approximately 1 month later in both 2012 and 2013 (Figs. 2*b* and 2*d*), indicating migration of most of the NL-population from the Nishinoko Lagoon to Lake Biwa.

The CPUE of wild juveniles in the Nishinoko Lagoon peaked in June in both 2012 and 2013, but rapidly decreased toward July and August (Figs. 2*a* and 2*c*). The CPUE of wild juveniles in the Chomeiji River also peaked in June 2012 and 2013, but decreased dramatically toward late June to August (Figs. 2*b* and 2*d*), indicating that most wild juveniles also migrated from the Nishinoko Lagoon to Lake Biwa.

Lake wide migration across Lake Biwa

SL and CF of fish at age 0 recaptured in the main lake in autumn (October–November) 2012 and 2013 ranged between 73.9 and 83.7 mm (Figs. 5*a* and 5*b*) and 18.1–20.5 (Figs. 5*c* and 5*d*), respectively, indicating an increase in SL and CF from the natal habitats in spring. In autumn 2012 and 2013, the ALC-marked honmoroko (NL-fish, IL-fish) at age 0 were distributed near-shore along both northern and southern parts, from the natal habitats (Figs. 3*a*–3*c*). The distribution of NL-fish and IL-fish overlapped, with captures in depths of 11–13 m (Figs. 3*b*, 3*c*, 5*e*, and 5*f*). In the best GAM, the probability of ALC-marked fish in autumn was explained solely by latitude, being highest in northern parts from the natal habitats with a sharp nonlinear decrease in both directions (Fig. 3*d*; Tables 2 and 3). Distance from the natal habitats to the recapture points in the main lake in autumn 2012 and 2013 ranged from 10.6 to 17.9 km (Table 1).

SL and CF of fish at age 0 recaptured in the main lake in winter (January and February) 2013 and 2014 ranged between 70.9–82.2 (Figs. 5*a* and 5*b*) and 16.6–18.1 (Figs. 5*c* and 5*d*), respectively, indicating no increase in SL and decrease in CF from autumn to winter within Lake Biwa. In winter 2013 and 2014, the ALC-marked honmoroko (NL-fish, IL-fish) at age 0 were distributed offshore in northern parts of the lake from the natal habitats (Figs. 4*a*–4*c*). The distribution of NL-fish and IL-fish partly overlapped, with captures between 50.9 and 64.6 m (Figs. 4*b*, 4*c*, 5*e*, and 5*f*). In the best GAM, the probability of ALC-marked fish in winter was also explained solely by latitude, being highest in northern parts from the natal habitats (Fig. 4*d*; Tables 3 and 4). No significant sex-biased migration was detected in the winter season, since sex was not included in the best models (Table 2). Distance from the natal habitats to the recapture points in the main lake in winter 2013 and 2014 ranged from 15.2 to 27.2 km (Table 1).

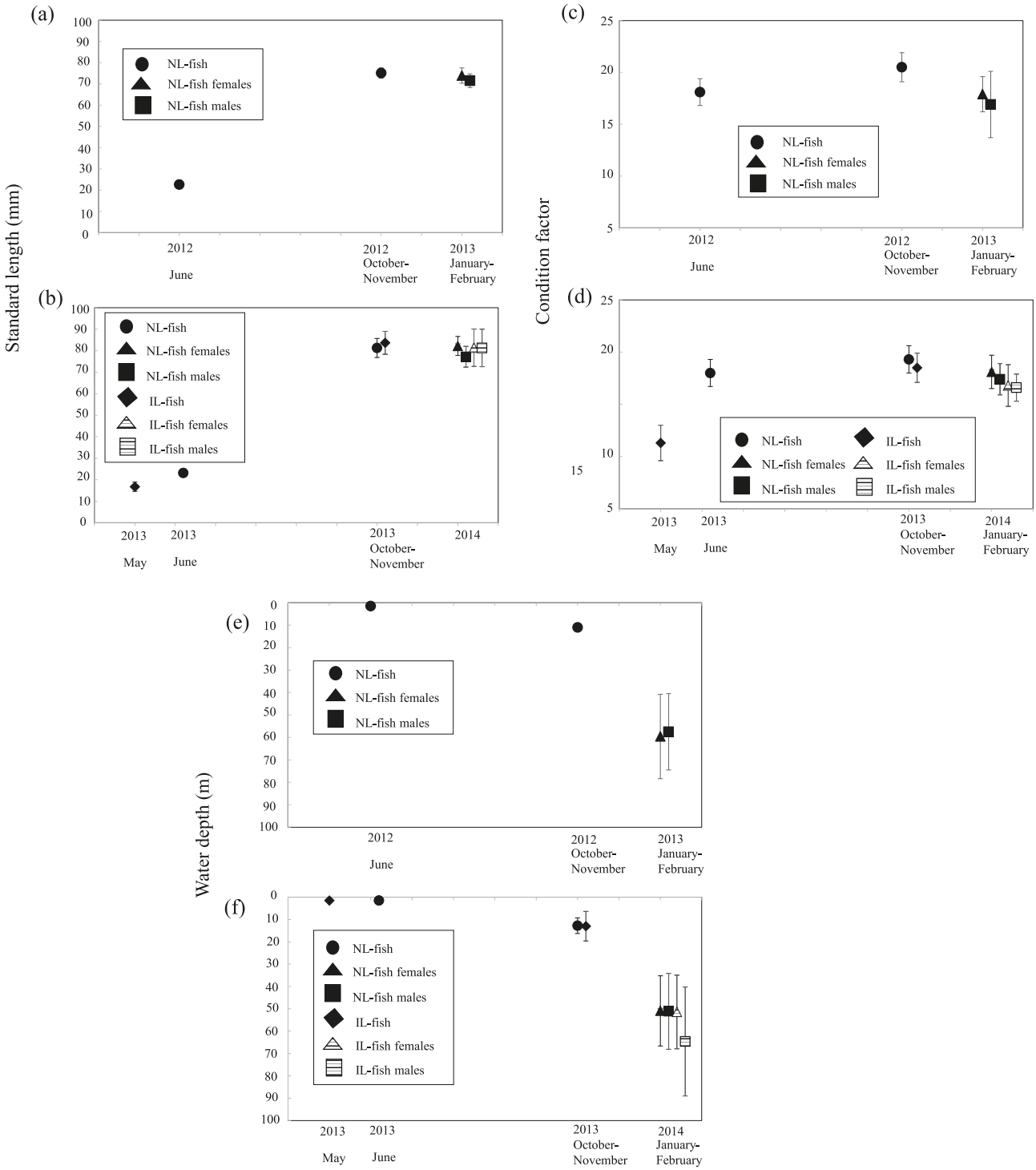
Natal homing

Captured ALC-marked fish (NL-fish, IL-fish) and wild fish around the spawning habitats from March to July 2013 and 2014 showed high GSI values, suggesting that they were mature spawning adults (Tables 1 and 5). The reproduction survey was conducted from March to July in 2013 and 2014, almost all of the ALC-marked fish released in Nishinoko (NL-fish) and Ibanaiko (IL-fish) lagoons were recaptured in the lagoon of their release (return rates 94.2%–98.9% in Nishinoko and 100% in Ibanaiko lagoons) (Fig. 6). The ratios of NL-fish captured around the lagoons and along the lakeshore differed significantly for both sexes among the spawning habitats (In 2013, females: $\chi^2 = 118.4$, $P < 0.001$; males: $\chi^2 = 47.1$, $P < 0.001$: In 2014, females: $\chi^2 = 322.5$, $P < 0.001$; males: $\chi^2 = 154.7$, $P < 0.001$), and were significantly higher (both sexes) in Nishinoko Lagoon than in the other habitats (Ryan multiple comparison tests, $P < 0.001$ for all comparisons). The ratios of IL-fish (both sexes) captured around the lagoons and along the lakeshore differed significantly among the spawning habitats (females: $\chi^2 = 14.1$, $P = 0.01$; males: $\chi^2 = 33.3$, $P < 0.001$). The ratios of male IL-fish were significantly higher in the Ibanaiko Lagoon (compared with the Nishinoko Lagoon), but not so among the other spawning habitats (Ryan multiple comparison tests: $P < 0.001$ for, Ibanaiko Lagoon vs. Nishinoko Lagoon, $P > 0.05$, for all other comparisons). The ratios of female IL-fish were not significantly higher among the spawning habitats (Ryan multiple comparison tests, $P > 0.05$, in all comparisons).

Discussion

Mass ALC marking of large numbers of fish produced by rice culture practices, together with the collaboration of com-

Fig. 5. Seasonal changes in standard length ((*a*) and (*b*)), condition factor ((*c*) and (*d*)), and water depth ((*e*) and (*f*)) (mean ± standard deviation) of alizarin complexone (ALC)-marked fish (NL-fish, IL-fish) at age 0 in Lake Biwa. (*a*), (*c*), and (*e*) stocked to Nishinoko Lagoon, June 2012: NL-fish. (*b*), (*d*), and (*f*) stocked to Nishinoko Lagoon, June 2013: NL-fish. Stocked to Ibanaiko Lagoon, May 2013: IL-fish.



mercial fishermen, was applied to evaluate migration of honmoroko during juvenile to adult stages. Honmoroko moved out of nursery lagoons each spring to enter Lake Biwa and then return to their natal lagoon to spawn the following spring. These results highlighted the importance of protecting individual spawning/nursery habitats, and maintaining connectivity between each spawning/nursery habitat and the lake system.

The rapid decline in densities of ALC-marked and wild juveniles in the Nishinoko Lagoon and Chomeiji river from spring

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Table 3. Results of best generalized additive models of migration pattern of honmoroko (*Gnathopogon caerelescens*) in Lake Biwa in autumn and winter.

Factors	Estimate	SE	*z*
***Autumn season* (Deviance explained = 13.9%)**			
(Intercept)	−3.44	0.15	
	EDF	Ref. Df	Chi.sq
Latitude	5.12	6.16	107.8***
***Winter season* (Deviance explained = 11.5%)**			
(Intercept)	−3.02	0.33	
	EDF	Ref. Df	Chi.sq
Latitude	2.98	3.70	271.9***

***indicate $p < 0.001$.

through summer strongly suggested that most honmoroko juveniles migrated from the nursery lagoon to an adult pelagic habitat in the large lake. Animal migrations are often caused by adaptations to spatial and temporal variations in resource availability (Borcherding et al. 2002). Zooplankton, such as rotifers, constitute an important food resource for juvenile honmoroko (Kikko et al. 2018*a*), the abundance of the former in the nursery lagoons adjacent to the lake generally increasing in May and June, but decreasing in July and August (Kikko et al. 2018*a*). On the other hand, zooplankton abundance in Lake Biwa usually increases from mid-June, maintaining a high level until early November (Yoshida et al. 2001). This difference in prey seasonal dynamics between the lagoon and lake habitats may have brought about the seasonal migratory patterns of juvenile honmoroko. Therefore, migration between the lagoon and lake habitats from spring through summer is thought to be feeding migration.

In autumn, the distribution of ALC-marked fish along the lake near-shore areas showed an increase in SL and CF from summer to autumn, suggested that they had migrated from the natal lagoons toward the near-shore areas in the main lake for feeding and growth. This is in keeping with zooplankton tending to be more abundant during summer and autumn in near-shore rather than pelagic areas in the main lake (Suzuki et al. 2019). In addition, while the overall decrease in food availability in the lake affects honmoroko movement seasonally, it can be expected that relative differences in food availability across the lake within the summer-fall period would also be an important factor driving movement patterns. In winter, the ALC-marked fish were distributed in offshore areas (having demonstrated no increase in SL and a decrease in CF from autumn to winter). Zooplankton becomes less abundant throughout Lake Biwa in winter (Yoshida et al. 2001). Honmoroko in Lake Biwa have already been found to not actively feed in winter, with stomachs being completely empty and no evidence of growth (Shiga Prefectural Fisheries Experimental Station 1942; Maki 1968). Many temperate fishes prefer lentic water conditions, especially in winter when low temperatures limit metabolism and swimming performance (Lucas and Baras 2001). Thus, in winter, available habitat for honmoroko in near-shore shallow areas in the north basin probably declined due to strong winds and changes in water temperature (Okamura 2008). On the other hand, in offshore areas, a huge deep zone characterized by stable water temperature and dissolved oxygen exists (annual water temperature average 7.7 °C at 80 m depth, annual dissolved oxygen average 8 mg/L, Okamura 2008). The relative differences in these habitats in winter facilitate fish movement to the deep off-shore areas, being a more stable environment in which to overwinter (Fujioka and Saegusa 2015).

In spite of apparent lake-wide migration in Lake Biwa, most ALC-marked fish with high GSI from March to June then appeared to return to each nursery lagoon. Eggs of honmoroko occurred in each lagoon from late March to mid-June, suggesting that these lagoons had been used as spawning habitats for honmoroko during that period (Kikko et al. 2015*c*, 2015*d*). Although we were unable to sample every spawning area with equal sampling effort, the current geographic coverage of honmoroko spawning areas sampled allowed an assessment of the homing intensities of the species. The results suggested that Homoroko migrate to each natal lagoon for spawning from offshore areas in winter, thereby showing natal homing. Similar natal homing has been found in some anadromous salmonids (Quinn 2005; Keefer and Caudill 2014). Natal homing has also been shown in the common carp (Banet et al. 2022), but no other cyprinid has been so quantified to date (L'bée-Lund and VØllestad 1985; Tynus 1990; Sungani et al. 2016). Hendry et al. (2004) predicted that homing of salmon and related species increases with increasing spatial variation and site quality. Spawning sites of honmoroko were distributed across almost the entire lakeshore, including the southern basin, until 1995 (Fujioka 2013; Mabuchi et al. 2022), although now restricted to the lagoons adjacent to the northern basin, and along the western and eastern shoreline habitats in the northern basin (Fig. 6, Kikko et al. 2014, 2015*d*). The lagoons provide suitable spawning and nursery grounds because Honmroko can spawn in both reed zones and lagoon inlets. The former are also highly productive, with a rich and stable zooplankton population in spring (Kikko et al. 2015*b*, 2018*a*, 2020). On the other hand, the western and eastern shoreline habitats are currently less suitable compared with the lagoons due to the loss of eggs due to water level manipulation and a great abundance of predators (Kikko et al. 2012, 2015*b*, 2017). Therefore, the current spatial variations in spawning habitat quality may have increased the propensity of natal homing of honmoroko to the lagoons.

Previous studies of cyprinids have reported long-distance migration among different habitats, and that suggested the importance of protecting large-scale habitat connectivity (Tynus 1990; Lucas and Batley 1996; Irving and Moddle 2000; Walters et al. 2014). The present study provided quantitative evidence to support such a management strategy by demonstrating juvenile migration and lake-wide movement of honmoroko within a large inland water system. Furthermore, the natal homing evident in this study suggests that degradation or loss of a few nursery habitats might result in a substantial decline of the entire honmoroko resource in the system. Many lagoons associated with Lake Biwa have already disappeared due to reclamation for agricultural land use, and cur-

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Table 4. Sampling effort for honmoroko collected around Lake Biwa spawning habitats during 2013 and 2014 spawning seasons.

Sampling location	Sampling season	Sampling days	Sampling gear
	2013		
Lakeshore (western area)	April–May	7	Backpack electrofishig
	May	8	Set net
Lakeshore (eastern area)	April–May	6	Backpack electrofishig
	June	1	Dip net
Ibanaiko Lagoon	March–April	10	Gill nets
	May	6	Bait fishing
	April–May	2	Dip net
Nishinoko Lagoon	April–June	4	Boat electrofishing
	April–June	10	Backpack electrofishig
	March–June	8	Set net
	March	1	Gill net
	2014		
Lakeshore (western area)	April–May	5	Backpack electrofishig
	May–June	28	Set net
Matsunoki Lagoon	April–May	3	Backpack electrofishig
Lakeshore (eastern area)	April–May	4	Backpack electrofishig
	June	2	Dip net
Sonenuma Lagoon	April	2	Boat electrofishing
Ibanaiko Lagoon	March–May	12	Gill net
	July	1	Backpack electrofishig
Nishinoko Lagoon	May–June	5	Boat electrofishing
	April–May	5	Backpack electrofishig
	March–June	8	Set net
	March	1	Gill net

Table 5. Size and gonadosomatic indices of wild honmoroko collected around Lake Biwa spawning habitats during the 2013 and 2014 spawning seasons.

Sampling location	Sampling season	No. of fish	Sex	Standard length (mm) mean ± SD	Gonadosomatic index (%) mean ± SD
	2013				
Lakeshore (western area)	April–May	267	Female	71.7 ± 5.1	10.4 ± 5.4
		334	Male	67.9 ± 5.2	6.2 ± 3.0
Lakeshore (eastern area)	April–June	122	Female	73.9 ± 3.5	13.7 ± 6.0
		222	Male	70.8 ± 3.6	6.1 ± 2.0
Ibanaiko Lagoon	March–May	428	Female	81.5 ± 4.1	11.6 ± 5.0
		307	Male	79.7 ± 4.0	7.8 ± 2.5
Nishinoko Lagoon	March–June	288	Female	82.9 ± 4.5	14.0 ± 7.3
		1535	Male	79.9 ± 4.3	5.7 ± 2.2
	2014				
Lakeshore (western area)	April–June	383	Female	82.0 ± 5.7	10.7 ± 5.2
		282	Male	76.8 ± 5.2	6.6 ± 2.6
Matsunoki Lagoon	April–May	54	Female	76.3 ± 5.3	12.9 ± 6.7
		63	Male	72.8 ± 6.2	6.7 ± 1.5
Lakeshore (eastern area)	April–June	63	Female	83.4 ± 4.7	17.7 ± 6.6
		81	Male	76.7 ± 5.9	6.8 ± 2.0
Sonenuma Lagoon	April	21	Female	89.9 ± 5.0	9.1 ± 4.4
		56	Male	84.3 ± 5.9	6.2 ± 1.4
Ibanaiko Lagoon	March–July	369	Female	91.3 ± 5.1	12.8 ± 3.7
		340	Male	87.6 ± 4.3	7.1 ± 4.0
Nishinoko Lagoon	March–June	84	Female	85.2 ± 8.1	10.6 ± 5.8
		305	Male	80.7 ± 6.9	6.9 ± 5.0

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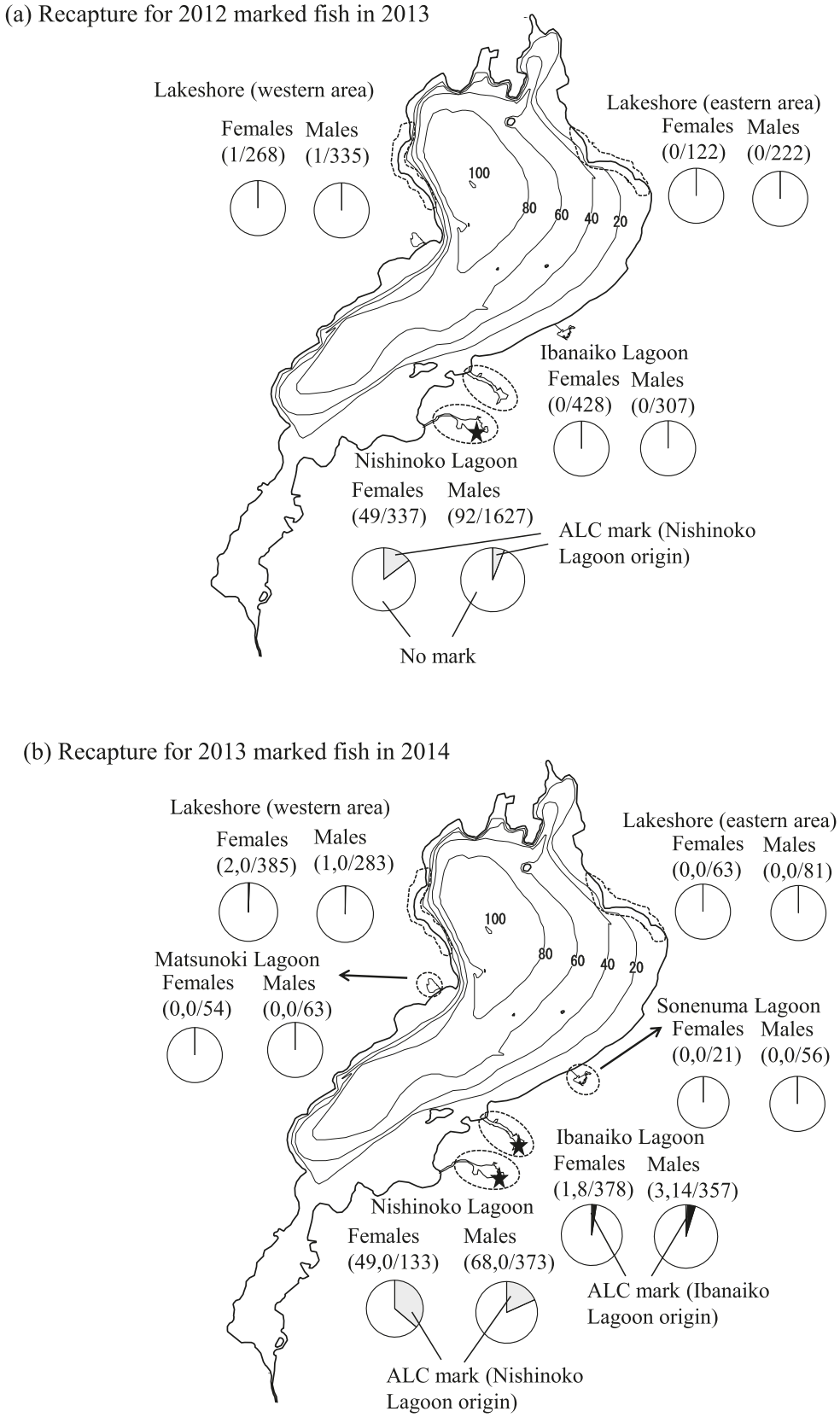
Fig. 6. Recaptures in the 2013 and 2014 spawning seasons (March–July) of alizarin complexone (ALC)-marked honmoroko stocked in Nishinoko Lagoon (NL-fish) in 2012 and 2013, and ALC dot-ring marked honmoroko stocked in Ibanaiko Lagoon (IL-fish) in 2013. Dashed areas indicate current major spawning areas and sampling locations. (*a*) Ratios of ALC-marked individuals (males and females) (NL-fish) in 2013 (ALC-marked individuals/total individuals caught in each spawning area). (*b*) Ratios of ALC small-ring marked individuals (males and females) (NL-fish) and ALC dot-ring marked individuals (males and females) (IL-fish) in 2014 (NL-fish, IL-fish/total individuals caught in each spawning area).



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rently extant lagoons are now highly fragmented by weirs connected to Lake Biwa and degraded (Shibata et al. 2011), even though such lagoons have valuable spawning and nursery grounds for honmoroko (Kikko et al. 2018*b*). This must include passage for adult spawning migration to the natal lagoons as well as juveniles migrating to the main lake from the lagoons. On the other hand, the distinct possibility that fish may simply not be dispersing far from their initial natal lagoons needs to be evaluated. Although most ALC-marked fish with high GSI were caught around the nursery lagoons, it could not be entirely ruled out that fish caught in the natal lagoons represented a few individuals that had never left the areas and/or may not have been sampled efficiently. Microchemical analyses of fish otoliths have considerable potential for revealing patterns of origin and movements (Gibson-Reinemer et al. 2009; Pracheil et al. 2014; Carlson et al. 2017), which would benefit our further understanding of the migration history, habitat use, and natal homing of honmoroko.

The extensive ALC mark-recapture programs, together with the collaboration of commercial fishermen, demonstrated that honmoroko in Lake Biwa: (1) migrate for feeding from a nursery lagoon toward the main lake and move along the near-shore from spring to autumn; (2) undergo refuge (wintering) migration from the near-shore shallow habitats to off-shore deep habitats from autumn to winter; and (3) exhibit natal homing from winter to spring. Of particular importance, the study demonstrated that a novel example of lake wide migration in a small cyprinid fish during its life-history. These findings have substantial implications for the interpretation and management of habitat connectivity during juvenile to adult stages. Furthermore, the study highlighted the substantial importance that protection of a few distant spawning/nursery habitats can have in achieving sustainable inland fisheries.

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Data availability

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Competing interests

The authors declare that they have no conflict of interest.

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